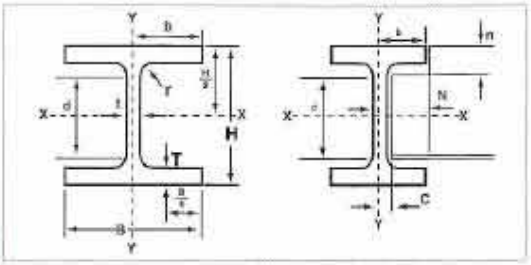
Dimensions and Properties



4c (i) H Bearing Piles (Imperial Units) - Part I

Designation				Height Of Bearing Pile	Width Of Bearing Pile	Thickness		Root Radius	Depth Between Fillet	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
Size H x B		Unit Weight				Flange	Web				Flange	Web	End Clearance	Notch	
				H	B	T	t	r	d	A	b/T	d/t	C	N	n
in	(mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm^2			mm	mm	mm
8 x 8	(203 x 203)	30	44.9	200.2	205.9	9.5	9.5	10.2	160.8	57.2	10.8	16.9	7	108	20
		36	53.9	204.0	207.7	11.4	11.3	10.2	160.8	68.7	9.11	14.2	8	108	22
10 x 10	(254 x 254)	42	63.0	247.1	256.6	10.7	10.6	12.7	200.3	80.2	12.0	18.9	7	133	23
		48	71.0	249.7	258.0	12	12	12.7	200.3	90.4	10.8	16.7	8	133	25
		57	85.1	254.3	260.4	14.3	14.4	12.7	200.3	108	9.10	13.9	9	133	27
12 x 12	(305 x 305)	53	78.9	299.3	306.4	11.1	11	15.2	246.7	100	13.8	22.4	8	158	26
		59	88.0	301.7	307.8	12.3	12.4	15.2	246.7	112	12.5	19.9	8	158	28
		63	93.0	303.3	308.0	13.1	13.1	15.2	246.7	119	11.8	18.8	9	157	28
		64	94.9	303.7	308.7	13.3	13.3	15.2	246.7	121	11.6	18.5	9	158	29
		74	110.0	307.9	310.7	15.4	15.3	15.2	246.7	140	10.1	16.1	10	158	31
		84	125.0	311.9	312.3	17.4	17.4	15.2	246.7	159	8.97	14.2	11	157	33
		85	126.1	312.3	312.9	17.6	17.5	15.2	246.7	161	8.89	14.1	11	158	33
		100	149.1	318.5	316.0	20.7	20.6	15.2	246.7	190	7.63	12.0	12	158	36
		121	180.0	326.7	319.7	24.8	24.8	15.2	246.7	229	6.45	9.95	14	157	40
		125	186.0	328.3	320.9	25.6	25.5	15.2	246.7	237	6.27	9.67	15	158	41
		149	222.9	337.9	325.7	30.4	30.3	15.2	246.7	284	5.36	8.14	17	158	46
13 x 13	(330 x 330)	60	89.3	318.5	327.7	11.68	11.68	15.2	264.7	113	14.0	22.7	8	168	27
		73	108.6	323.8	330.3	14.35	14.35	15.2	264.7	139	11.5	18.4	9	168	30
		87	129.5	328.9	332.9	16.89	16.89	15.2	264.7	164	9.85	15.7	10	168	32
		100	148.8	334.0	335.4	19.43	19.43	15.2	264.7	190	8.63	13.6	12	168	35
14 x 14 1/2	(356 x 368)	73	108.9	346.4	371.0	12.9	12.8	15.2	290.2	139	14.4	22.7	8	189	28
		89	133.0	352.0	373.8	15.7	15.6	15.2	290.2	169	11.9	18.6	10	189	31
		102	152.0	356.4	376.0	17.9	17.8	15.2	290.2	194	10.5	16.3	11	189	33
		117	173.9	361.4	378.5	20.4	20.3	15.2	290.2	221	9.28	14.3	12	189	36
		121	180.0	362.9	378.8	21.1	21.1	15.2	290.3	230	8.98	13.8	13	189	36

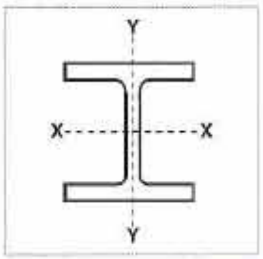
Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.

4c (i) Bearing Piles (Metric Units) - Part A

Designation		Height Of Bearing Pile	Width Of Bearing Pile	Thickness		Root Radius	Depth Between Fillet	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
Size H x B	Unit Weight			Flange	Web				Flange	Web	End Clearance	Notch	
		H	B	T	t	r	d	A	b/T	d/t	C	N	n
mm	kg/m	mm	mm	mm	mm	mm	mm	cm^2			mm	mm	mm
200 x 200	56.2	200	204	12	12	13	150	71.5	8.50	12.5	8	106	25
250 x 250	63.8	244	252	11	11	13	196	81.3	11.5	17.8	8	131	24
	64.4	244	252	11	11	16	190	82.1	11.5	17.3	8	131	27
	82.2	250	255	14	14	16	190	105	9.11	13.6	9	131	30
300 x 300	83.4	294	302	12	12	13	244	106	12.6	20.3	8	155	25
	84.5	294	302	12	12	18	234	108	12.6	19.5	8	155	30
	106	300	305	15	15	18	234	135	10.2	15.6	10	155	33
	142	310	310	20	20	18	234	181	7.75	11.7	12	155	38
350 x 350	105	338	351	13	13	13	286	133	13.5	22.0	9	179	26
	106	338	351	13	13	20	272	135	13.5	20.9	9	179	33
	129	344	354	16	16	13	286	165	11.1	17.9	10	179	29
	131	344	354	16	16	20	272	167	11.1	17.0	10	179	36
	154	350	357	19	19	13	286	196	9.39	15.1	12	179	32
	156	350	357	19	19	20	272	198	9.39	14.3	12	179	39
400 x 400	140	388	402	15	15	22	314	178	13.4	20.9	10	204	37
	168	394	405	18	18	22	314	214	11.3	17.4	11	204	40
	197	400	408	21	21	22	314	251	9.71	15.0	13	204	43
	235	408	412	25	25	22	314	300	8.24	12.6	15	204	47

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.

Dimensions and Properties



4c (i) H Bearing Piles (Imperial Units) - Part II

Surface Area Per metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation			
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y					Unit Weight		Size H x B	
m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	u	x	H	J	lb/ft	kg/m	(mm)	in
1.19	4100	1384	8.46	4.92	410	134	459	206	0.827	18.6	0.126	19.2	30	44.9	(203 x 203)	8 x 8
1.20	5027	1705	8.55	4.98	493	164	557	252	0.827	15.8	0.158	32.7	36	53.9		
1.48	8861	3016	10.5	6.13	717	235	799	360	0.827	20.5	0.421	34.3	42	63.0	(254 x 254)	10 x 10
1.49	10070	3439	10.6	6.17	807	267	904	409	0.826	18.4	0.486	48.4	48	71.0		
1.50	12290	4216	10.6	6.24	966	324	1092	498	0.825	15.6	0.607	81.8	57	85.1		
1.78	16450	5327	12.8	7.28	1099	348	1218	531	0.832	23.9	1.11	46.9	53	78.9	(305 x 305)	12 x 12
1.78	18430	5985	12.8	7.31	1221	389	1360	595	0.830	21.6	1.25	64.2	59	88.0		
1.79	19690	6387	12.9	7.33	1298	415	1449	635	0.831	20.5	1.34	76.4	63	93.0		
1.79	20050	6529	12.9	7.35	1320	423	1474	648	0.830	20.2	1.38	80.0	64	94.9		
1.80	23560	7710	13.0	7.42	1531	496	1720	762	0.830	17.7	1.65	122	74	110.0		
1.81	27040	8849	13.0	7.46	1734	567	1961	872	0.829	15.8	1.92	177	84	125.0		
1.82	27410	9002	13.1	7.49	1755	575	1986	885	0.829	15.7	1.95	182	85	126.1		
1.83	33070	10910	13.2	7.58	2076	691	2370	1066	0.828	13.5	2.42	295	100	149.1		
1.86	40970	13550	13.4	7.69	2508	847	2897	1313	0.827	11.5	3.09	510	121	180.0		
1.86	42610	14140	13.4	7.73	2596	881	3003	1366	0.827	11.1	3.24	560	125	186.0		
1.89	52700	17580	13.6	7.87	3119	1079	3653	1680	0.826	9.55	4.16	943	149	222.9		
1.90	20940	6857	13.6	7.79	1315	418	1457	639	0.830	24.3	1.61	57.9	60	89.3	(330 x 330)	13 x 13
1.91	26200	8628	13.7	7.88	1618	522	1808	800	0.829	20.1	2.07	106	73	108.6		
1.93	31430	10400	13.8	7.96	1911	625	2151	959	0.828	17.3	2.53	171	87	129.5		
1.94	36860	12240	13.9	8.03	2207	730	2502	1123	0.828	15.2	3.03	260	100	148.8		
2.13	30630	10990	14.9	8.90	1769	592	1956	903	0.823	24.2	3.06	84.6	73	108.9	(356 x 368)	14 x 14 1/2
2.14	37980	13680	15.0	8.99	2158	732	2406	1119	0.822	20.1	3.87	151	89	133.0		
2.16	43970	15880	15.1	9.05	2468	845	2767	1293	0.821	17.8	4.55	223	102	152.0		
2.17	51010	18460	15.2	9.13	2823	976	3186	1497	0.821	15.8	5.37	330	117	173.9		
2.17	53040	19140	15.2	9.13	2923	1011	3306	1552	0.821	15.3	5.59	367	121	180.0		

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.

4c (i) Bearing Piles (Metric Units) - Part B

Surface Area Per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation	
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y					Unit Weight	Size H x B
m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3	u	x	H	J	kg/m	mm
1.17	4983	1702	8.35	4.88	498	167	565	257	0.826	14.3	0.150	39.6	56.2	200 x 200
1.45	8703	2938	10.3	6.01	713	233	797	357	0.828	19.5	0.399	37.1	63.8	250 x 250
1.45	8787	2939	10.3	5.98	720	233	805	358	0.829	19.0	0.399	39.5	64.4	
1.46	11480	3877	10.5	6.09	919	304	1039	468	0.827	15.4	0.540	79.0	82.2	
1.75	16640	5514	12.5	7.20	1132	365	1260	558	0.829	21.9	1.10	56.3	83.4	300 x 300
1.74	16870	5517	12.5	7.16	1147	365	1277	560	0.831	21.1	1.10	61.4	84.5	
1.76	21540	7106	12.6	7.26	1436	466	1614	716	0.829	17.4	1.44	116	106	
1.79	29870	9955	12.9	7.42	1927	642	2199	992	0.828	13.4	2.09	271	142	
2.03	27740	9377	14.4	8.39	1642	534	1822	815	0.827	23.5	2.48	81.4	105	350 x 350
2.02	28190	9381	14.4	8.33	1668	535	1851	818	0.829	22.5	2.48	90.3	106	
2.05	34880	11840	14.6	8.48	2028	669	2269	1024	0.826	19.4	3.19	151	129	
2.04	35330	11850	14.6	8.43	2054	669	2299	1027	0.828	18.7	3.19	164	131	
2.07	42350	14430	14.7	8.57	2420	808	2730	1241	0.826	16.5	3.95	252	154	
2.06	42800	14430	14.7	8.53	2446	809	2760	1244	0.827	16.1	3.95	270	156	
2.32	48970	16260	16.6	9.55	2524	809	2802	1237	0.830	22.5	5.66	156	140	400 x 400
2.33	59720	19960	16.7	9.65	3031	986	3390	1511	0.828	19.2	7.05	264	168	
2.35	70890	23810	16.8	9.75	3545	1167	3992	1794	0.828	16.7	8.55	415	197	
2.38	86470	29200	17.0	9.87	4239	1418	4818	2185	0.827	14.2	10.7	694	235	

Note: Calculated based on 1 kg = 2.2046 lb and 1 m = 3.2808 feet.